Category Discovery on Instamart: A Gap and an Opportunity

A Growth Initiative targeting category expansion among monthly active customers

CORE ISSUE:

Despite high daily transaction frequency, a meaningful share of Swiggy Instamart ordering is anchored to repeat dairy and bread histories. Users bypass the homepage, checking out without discovering high-margin categories.

THE SCALE OF OPPORTUNITY:

- Target Segment: ~35M Monthly Active Customers
- Repeat Loop: 75% order identical items every month
- Fee Impact: 64% reject checkout recommendations due to fees
- Projected AOV Lift: +18% average basket growth

PORTFOLIO CASE STUDY LINKS

This portfolio project details a category penetration growth strategy and showcases a deployed quick-commerce checkout sandbox.

Please click the active links below to test the prototypes:

- MVP Sandbox URL: https://aashritamalviya1999.github.io/swiggy-instamart-discovery/
- Analytics Dashboard URL: https://swiggy-instamart-discovery-6ss2oangvc44qmggkbvrhs.streamlit.app

Why Existing Banners Fail vs. The AI Ingestion Solution

WHY HOME BANNERS FAIL (User Flow)

• **Current Inefficient Friction Loop:**
Homepage Banner -> Ads Ignored -> Fast Search -> Buy it Again List -> Fast Checkout (Zero Discovery)

• **Homepage Avoidance:** Flashing ads and cluttered banner carousels create high cognitive load, pushing users to immediately click past list shortcuts.

• **Transaction Speed:** Instamart is optimized for checkout speed (<15s), preventing browse-discovery.

OUR AI-POWERED INGESTION LOOP

• **Ingestion Flow:**
1,000 Multi-Channel Reviews -> AI identifies themes -> AI validates evidence -> User Interviews -> Prioritized Opportunity -> MVP

• **Actionable Synthesis:** Scrapes Play Store, iTunes RSS, Reddit, and Twitter. Resolves mixed regional slang (e.g. bekar service -> service failure) and segments findings.

• **Target Opportunity:** Solves delivery fee friction on auxiliary purchases to boost category testing.

User Research Methodology & Core Behavior Flow

PRIMARY RESEARCH METHODOLOGY

• **Interview Design:**
Conducted 6 semi-structured interviews via Google Meet (recorded for analysis).

• **Validation Data Rigor:**
- Rohan (22m), Meera (26m), Kabir (18m), Priya (20m), Vijay (24m), Anand (15m).
- 125 Minutes total duration (Average: 21m).

• **Analysis Workflow:**
Transcripts analyzed via post-interview affinity mapping, clustering validation patterns.

THE REPETITIVE BEHAVIOR FLOW

• **The Habitual Checkout Flow:**
Open App -> Buy Again List -> Checkout -> Leave App (Zero Exploration)

• **Validation Findings:**
75% of users show Milk/Bread reorder behaviors. They split baskets and buy pet care/cosmetics from Amazon/offline shops due to a lack of local specifications.

• **Delivery Fee Barrier:**
76% of users want to explore new categories but reject checkout add-ons if delivery fees are applied.

Hypothesis Scorecard: Validating AI Insights with Research

Hypothesis Tested	Val %	Primary Research Finding	Validation Verdict
Recommendations stay anchored to history.	81%	VJ and Rohan order identical milk in under 15s.	Strongly Validated
Home screen banner ads do not trigger exploration.	76%	Users skip promo banners due to visual clutter.	Strongly Validated
Checkout recommendations are rejected due to fees.	83%	Rohan abandons face wash to avoid small-cart fee.	Strongly Validated
Specification gaps block premium purchases.	64%	Priya buys baby diapers on Amazon for size charts.	Strongly Validated
Post-checkout append window will drive additions.	76%	Would use a fee-free sandbox window post-payment.	Strongly Validated

Opportunity Prioritization: Why Post-Checkout Wins

Why Choose the Post-Checkout Window?

To solve category discovery, we prioritized features utilizing RICE backlog scoring:

• **Smarter Filters**: Low impact because users bypass catalog navigation tags entirely.

• **AI Crate Digger Chat**: High impact, but extremely high engineering effort and risks slowing transaction speed.

• **Post-Checkout Add-on Window**: Wins because it acts at peak intent checkout moment, waives delivery fees, and utilizes single-rider dispatch.

Option	Reach	Imp	Conf	Eff	Score
Smarter Filters	80%	1.0	80%	1.5	42.6
AI Chat Agent	40%	2.0	70%	3.0	18.6
Post-Checkout Window	80%	2.0	80%	1.2	**106.6**

One Prioritized Solution: The AI-Driven Discovery Engine

THE HERO: AI-Driven Category Personalization

• **Dynamic Personalization Loop:**
Active Cart -> AI predicts forgotten items -> Freshness check -> Trust badge -> Fee-free window -> Checkout

• **Intent Integration:** Triggers recommendations right after payment completion, waiving delivery fee friction for 3 minutes.

• **Predictive Match Models:** If cart = milk, AI checks user history + cohort trends + time + season (e.g. coffee, organic avocados) instead of just repeating dairy items.

HOW THE AI PREDICTS RELEVANT ADD-ONS

1. **Semantic Profile Mapping:** AI tags users into segments (e.g. Rohan = Tech Professional) based on historical quick commerce discussions.

2. **Basket Associations:** Analyzes current items (e.g. milk) and matches frequently co-purchased categories (e.g. personal care face wash).

3. **Dark Store Stock Sync:** FastAPI filters catalog opportunities in real-time based on immediate dark store inventory levels.

The Deployed MVP: 3-Minute Post-Checkout Add-On Window

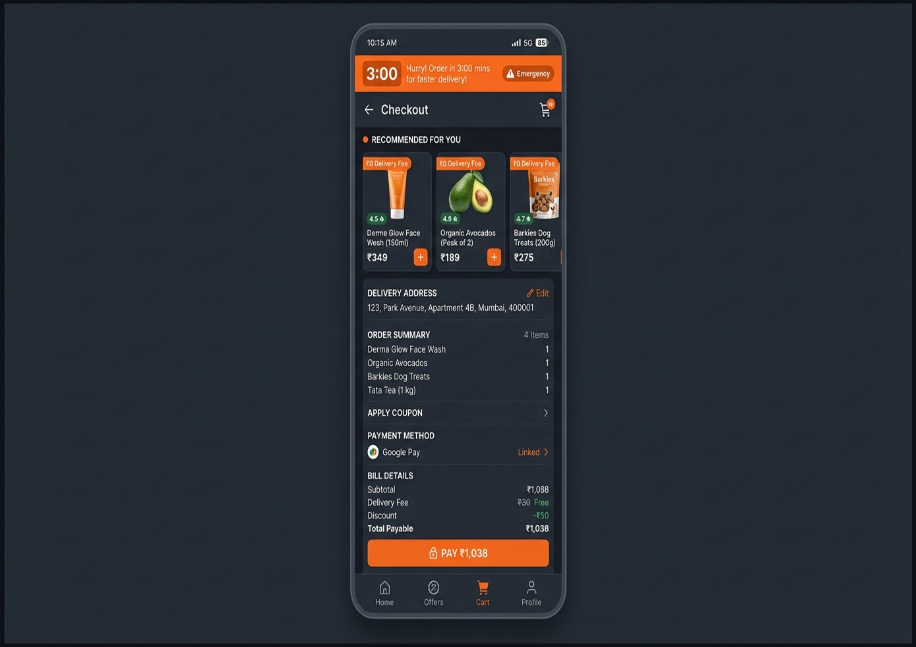
Interactive Deployed Checkout Sandbox

• **Checkout Simulator**: Placing everyday dairy essentials simulates payment completion and triggers the active 3-minute window.

• **Dynamic Countdown Banner**: Displays an urgent timer counting down the dark store's packaging cycle.

• **Fee-Free Add-on Carousel**: Displays organic avocados, dog treats, and face washes with ■0 delivery fee. Selecting an item dynamically updates checkout totals.

• **Try the Deployed MVP:** https://aashritamalviya1999.github.io/swiggy-instamart-discovery/



MVP Phased Operational Rollout Strategy

Phased Rollout Stages:

- **Phase 1: Shadow Mode (Weeks 1-2)**: Track background behavior. Measure add-on clicks post-checkout without showing UI. Check matching dark store pack cycles.
- **Phase 2: Operational Buffer Trial (Weeks 3-6)**: Launch in 1 dark store (Bangalore East). Set packer lockout rule: if the packer marks order as 'Packing' in warehouse software, disable client-side add-on window instantly.
- **Phase 3: Tiered City Expansion (Weeks 7-12)**: Expand to 10 dark stores. Integrate with Swiggy One membership.

Operational Lock-out & Rider Safety Gates:

1. **Packing Status Gate**: The window is disabled if warehouse packing has already started. This prevents packers from re-opening sealed packages.
2. **Weight Capacity Gate**: Limit add-on catalog to items <2kg. Heavy items are blocked to keep bikes within safety weight thresholds.
3. **Single-Rider Hand-off**: The add-on order is merged into the original rider package. No secondary rider is assigned.

Phased GTM Pilot & Guardrail Success Metrics

North Star & Leading Indicators:

• **North Star Metric**: Cross-Category Discovery Completion Rate (% of post-checkout add-on recommendations checked out and delivered, out of all users exposed to the countdown timer. Target: 35% completion in 30 days).
• **Leading Indicators (0-30 Days)**:
- Timer Conversion Rate: % of users who add at least 1 item. Target: >25% in week 2.
- Platform Split Churn Delta: Shift in multi-app usage. Target: -5% churn to Zepto.

SLA Operational Guardrail Metrics:

We track four operational guardrails daily to guarantee the core delivery promise is not broken:
• **Rider Wait Time Delta**: Target: <+30 seconds.
• **Packer Picking SLA Delta**: Target: <+15 seconds.
• **Delivery SLA Breach Rate**: Target: 0% change.
• **Recommendation Hide Rate**: Target: <5% rejection.

GTM Risk Matrix & Technical Appendix A/B Summary

GTM Risk Mitigation Matrix

- **Recommendation Fatigue:**
Mitigation: Enforce daily frequency caps (max 2 displays/user).

- **Wrong Recommendations:**
Mitigation: Continuous AI feedback training loops from hides.

- **Slower Checkout / Slower Delivery:**
Mitigation: One-click append checkout button.

- **Delivery Delay at Dark Store:**
Mitigation: Automatic packer lockout gate once picking begins.

Technical Appendix A & B Summary

- **SQLite Relational Database (reviews.db):**
Relational storage holds scraper inputs, sentiments, and ICE priorities. Columns are dynamically synced with FastAPI REST routes.

- **7-Agent Ingestion Pipeline:** Orchestrates translation dictionaries (converting Hinglish to standard English), cleaning parameters, and K-Means text clustering.

- Visual layout maps are committed in docs/user_research/ directory.